

Multiple Variable Calculus

2. The Jacobian in Three Dimensions

$$I = \iiint_V f(x, y, z) dx dy dz$$

$$x = x(u, v, \omega)$$

$$y = y(u, v, \omega)$$

$$z = z(u, v, \omega)$$

$$J(u, v, \omega) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial \omega} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial \omega} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial \omega} \end{vmatrix}$$

$$I = \iiint f(u, v, \omega) |J(u, v, \omega)| du dv d\omega, \quad dx dy dz \rightarrow |J(u, v, \omega)| du dv d\omega$$